

CREDIT RISK EARLY WARNING SYSTEM

Analyst Brief | Albee Manoj Thomas | SKEMA Business School | June 2026

EXECUTIVE SUMMARY

This analysis examined 30,000 retail credit card clients to identify borrowers showing early signs of default risk before it materialises. Using six months of payment behaviour, credit utilisation, and demographic data, clients were segmented into risk tiers to surface where portfolio exposure is most concentrated and where intervention will have the greatest impact.

30,000	6,636	22.12%	5.02bn
Total Clients	Total Defaults	Overall Default Rate	Total Portfolio Exposure

KEY FINDINGS

01 — Default Concentration

High-risk clients represent only 8.6% of the portfolio yet carry a 64.05% default rate — 5.5x higher than low-risk clients at 11.71%. Despite this extreme default rate, high-risk clients hold only 4.72% of total exposure (237M of 5.02bn), meaning the volume risk sits in the medium tier which holds 21% of portfolio exposure at a 35.34% default rate.

Risk Tier	Clients	Default Rate	Total Exposure	% of Portfolio
risk_tier_high	2,590	64.05%	237,070,000	4.72%
risk_tier_medium	7,479	35.34%	1,055,192,000	21.00%
risk_tier_low	19,931	11.71%	3,732,267,680	74.28%

02 — Demographic Risk Patterns

Clients aged 60+ carry the highest default rate at 28.32%. High School educated clients default at 25.16% vs 19.23% for Graduate clients. The single highest-risk cohort is high-tier clients aged 50-59 with Graduate education — 83.33% default rate across 42 clients. The second highest-risk cohort is high-tier clients aged 40-49 with University education at 71.20% across 250 clients.

03 — Credit Limit Risk Concentration

Clients with credit limits below 50K carry the highest default rate at 31.79% across 7,676 clients. Risk decreases consistently as credit limit increases, with above-500K clients defaulting at only 11.17%. This inverse relationship suggests lower-limit clients face greater financial stress.

RECOMMENDATIONS

01	Prioritise outreach to high-risk tier clients immediately 2,590 clients carry a 64% default rate. A targeted early intervention programme — payment restructuring or credit counselling — focused on this segment could materially reduce default losses before they crystallise.
02	Apply stricter credit controls to below-50K limit segment This bucket has the highest default rate (31.79%) across 7,676 clients. Tightening approval criteria and reducing credit limits for new applicants in this band would lower portfolio-level default concentration.

Introduce age-adjusted monitoring for 60+ clients

Clients aged 60+ show the highest default rate at 28.32%. A dedicated monitoring flag for this cohort with proactive account reviews would give the risk team earlier visibility into deteriorating repayment behaviour.

LIMITATIONS & ASSUMPTIONS

Dataset reflects Taiwan credit card clients from 2005 — patterns may differ across modern banking markets. This is a descriptive segmentation analysis, not a predictive model — findings indicate elevated risk behaviour, not a probability score. Credit limit and payment history are used as proxies for financial health. External factors such as income and employment are not available.

Dataset: UCI Credit Card Default | 30,000
clients | 25 variables

Tools: MySQL · Python (pandas, numpy) ·
Power BI

[github.com/albee-thomas/credit-risk-early-war
ning](https://github.com/albee-thomas/credit-risk-early-warning)